# Scalability & Future Plan

| | |
|---|---|
| **Date** | 29 september 2026 |
| **Team ID** | 6ab2abc01266c775575a9dda |
| **Project Name** | Farm Management System |
| **Maximum Marks** | 1 Mark |

### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state, ensuring long-term viability and impact.

### Current System Limitations:

| S.No | Limitation | Impact | Priority to Address (High / Medium / Low) |
|---|---|---|---|
| 1 | Limited number of users can be handled initially | More users may affect system performance | High |
| 2 | Limited data storage and reporting features | Large farm records may become difficult to manage | Medium |
| 3 | Basic security and notification features | User data protection and timely alerts can be improved | High |

### Scalability Plan:

| S.No | Scalability Aspect | Current State | Proposed Upgrade / Solution |
|---|---|---|---|
| 1 | User Load | System is designed for a limited number of users. | Use a scalable server and database to support more farmers and users simultaneously. |
| 2 | Data Storage | Farm, crop and activity data are stored in a structured database. | Increase database capacity and use cloud storage for handling large amounts of farm data. |

| | | | |
|---|---|---|---|
| 3 | Performance | System works efficiently with the current number of users and records. | Optimize database queries and server performance to support more users and data. |
| 4 | Security | Basic login and user authentication are provided. | Implement stronger authentication, password encryption, role-based access and regular data backup. |

**Future Roadmap:**

| Phase | Planned Feature / Enhancement | Target Timeline | Expected Impact |
|---|---|---|---|
| Phase 2 | Mobile application and notification system | 3–6 Months | Farmers can access the system easily and receive timely alerts. |
| Phase 3 | Weather integration and smart irrigation support | 6–12 Months | Helps farmers make better decisions about irrigation and crop activities. |
| Phase 4 | AI-based crop recommendations and analytics | 12–18 Months | Provides useful insights for crop planning and improves farm management. |